

LI WEI (NEWAY) 李巍

weili3@nus.edu.sg

Assistant Professor ◊ School of Computing ◊ National University of Singapore ◊ wadmes.github.io/cv/

APPOINTMENTS

National University of Singapore, Singapore Assistant Professor, School of Computing	July 2026 – Present
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EDUCATION

Carnegie Mellon University, PA, USA Doctor of Philosophy, Department of Electrical and Computer Engineering Advisor: Professor Shawn Blanton & Professor José Moura	Aug 2021 – May 2026
The Chinese University of Hong Kong, Hong Kong Master of Philosophy, Department of Computer Science and Engineering Thesis: <i>Irregular Deep Data Embedding and Learning</i> (Faculty Outstanding Thesis Award) Supervisor: Professor Bei Yu	2019 – 2021
Bachelor of Science, ELITE Stream , Department of Computer Science and Engineering	2014 – 2018

SELECTED AWARDS AND HONORS

Fellowships & Scholarships

Croucher Fellowship <i>Approx. \$200,000; 2-year stipend and research grant support. The most prestigious award for PhDs from HK.</i>	Croucher Foundation	2026
Qualcomm Innovation Fellowship <i>\$100,000 research grant; awarded for high-impact innovation globally in EDA.</i>	Qualcomm Inc.	2024
Apple PhD fellowship in Integrated Systems <i>Approx. \$100,000; for elite IC PhD at designated premier universities; includes full funding and Apple mentorship.</i>	Apple Inc.	2024
Apple PhD fellowship in Integrated Systems <i>Approx. \$100,000; for elite IC PhD at designated premier universities; includes full funding and Apple mentorship.</i>	Apple Inc.	2022
Jack and Mildred Bowers Scholarship	CMU	2025
Dean's Fellowship	CMU	2021
Talent Development Scholarship	HKSAR Government	2021
Full Postgraduate Studentship	CUHK	2019-2021
ELITE Stream Student Scholarship	Faculty of Engineering, CUHK	2018
Undergraduate Admission Scholarship <i>¥200,000; Top 0.1% in National College Entrance Examination (Gaokao)</i>	Soong Ching Ling Foundation	2015-2018

Awards & Honors

ML and Systems Rising Star	MLCommons	2026
Best Paper Award, Honorable Mention <i>Invited for internal sharing at Apple (100+ attendees).</i>	ICLAD, top 3 out of 94 submissions	2026
Best Paper Award	ASP-DAC, top 2 out of 368 submissions	2021
Best Student Paper Award	ICTAI, top 1 out of 458 submissions	2019
Distinguished Paper Award	ISSTA, top 3 out of 142 submissions	2019
Faculty Outstanding Thesis Award <i>Sole recipient of the best thesis award per year in the Faculty of Engineering..</i>	Engineering Faculty, CUHK	2021
1st Place Award in EDA elite challenge	Chinese Institute of Electronics	2020
Richard Newton Young Student Fellow	DAC	2020
2nd Place Award in CAD Contest	ICCAD	2018

PATENTS

- [P2] **Wei Li**, Rongjian Liang, Haoxing Ren. “Differentiable Global Router”. U.S. Patent Application 18/965,415, Filed Dec 2, 2024. Assigned to NVIDIA Corp. - Patent Pending.
- [P1] **Wei Li**, Ronald Blanton, Jose M. F. Moura. “System and Method for Global Floorplanning via Semidefinite Programming”. U.S. Patent Application 18/910,938, Filed: Oct 9, 2024 —Patent Pending.

PUBLICATIONS

Journal Papers

- [J2] **Wei Li**, Yuzhe Ma, Yibo Lin, Bei Yu, “Adaptive Layout Decomposition with Graph Embedding Neural Networks”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**).
- [J1] **Wei Li**, Yuzhe Ma, Qi Sun, Zhang Lu, Yibo Lin, Iris Hui-Ru Jiang, Bei Yu, David Z. Pan, “OpenMPL: An Open Source Layout Decomposer”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**).

Conference Papers

- [C18] **Wei Li**, Yang Zou, Yixin Liang, Shawn Blanton and José Moura, “DEFT: Differentiable Framework for Circuit Testing”, IEEE International Test Conference (**ITC**), 2026.
- [C17] **Wei Li**, Yang Zou, Christopher Ellis, Ruben Purdy, Shawn Blanton and José Moura, “BRIDGE: Bridging Graph and Large Language Models in EDA”, IEEE International Conference on LLM-Aided Design (**ICLAD**), 2025. (**Best Paper Award, Honorable Mention**)
- [C16] Chris Nigh, Ruben Purdy, **Wei Li**, Subhasish Mitra, R.D. Blanton, “ IC-PEPR: PEPR Testing Goes Intra-Cell”, IEEE International Test Conference (**ITC**), 2025.
- [C15] Ruben Purdy, Chris Nigh, **Wei Li**, R.D. Blanton, “CHEF: CHaracterizing Elusive Logic Circuit Failure” , IEEE VLSI Test Symposium (**VTS**) 2025.
- [C14] Chris Nigh, Ruben Purdy, **Wei Li**, Subhasish Mitra, R.D. Blanton, “Faulty Function Extraction for Defective Circuits” , IEEE European Test Symposium (**ETS**) 2024.
- [C13] **Wei Li**, Rongjian Liang, Anthony Agnesina, Haoyu Yang, Chia-Tung Ho, Anand Rajaram, Haoxing Ren, “DGR: Differentiable Global Routing”, ACM/IEEE Design Automation Conference (**DAC**), San Francisco, 2024.
- [C12] **Wei Li**, Ruben Purdy, Jose Moura, Shawn Blanton, “Characterize the ability of GNNs in attacking logic locking”, ACM/IEEE Workshop on Machine Learning for CAD (**MLCAD**), Snowbird, Utah, Sep. 11–13, 2023.
- [C11] **Wei Li**, Fangzhou Wang, Jose Moura, Shawn Blanton, “Global floorplanning via semidefinite programming”, ACM/IEEE Design Automation Conference (**DAC**), San Francisco, July 9-13, 2023.
- [C10] **Wei Li**, Chris Nigh, Danielle Duvalsaint, Subhasish Mitra, R.D. Blanton, “PEPR: Pseudo-Exhaustive Physical Region Testing”, IEEE International Test Conference (**ITC**), Sep. 25 - Sep. 30, 2022.
- [C9] **Wei Li**, Guojin Chen, Haoyu Yang, Ran Chen, Bei Yu, “Learning Point Clouds in EDA”, ACM International Symposium on Physical Design (**ISPD**), Mar. 21–Mar. 24, 2021.
- [C8] **Wei Li**, Yuxiao Qu, Gengjie Chen, Yuzhe Ma, Bei Yu, “TreeNet: Deep Point Cloud Embedding for Routing Tree Construction”, IEEE/ACM Asian and South Pacific Design Automation Conference (**ASP-DAC**), Tokyo, Jan. 18–21, 2021. (**Best Paper Award**)
- [C7] **Wei Li**, Jialu Xia, Yuzhe Ma, Jialu Li, Yibo Lin, Bei Yu, “Adaptive Layout Decomposition with Graph Embedding Neural Networks”, ACM/IEEE Design Automation Conference (**DAC**), San Francisco, July 19–23, 2020.
- [C6] Husheng Zhou, **Wei Li**, Yuankun Zhu, Yuqun Zhang, Bei Yu, Lingming Zhang, Cong Liu, “DeepBillboard: Systematic Physical-World Testing of Autonomous Driving Systems”, ACM/IEEE International Conference on Software Engineering (**ICSE**), Seoul, May 23–29, 2020.
- [C5] Yuzhe Ma, Zhuolun He, **Wei Li**, Tinghuan Chen, Lu Zhang, Bei Yu, “Understanding Graphs in EDA: From Shallow to Deep Learning”, ACM International Symposium on Physical Design (**ISPD**), Taipei, Mar. 25–Apr. 01, 2020.

- [C4] Yuzhe Ma, Ran Chen, **Wei Li**, Fanhua Shang, Wenjian Yu, Minsik Cho, Bei Yu, “A Unified Approximation Framework for Deep Neural Networks”, The IEEE International Conference on Tools with Artificial Intelligence (ICTAI) 2019. **(Best Student Paper Award)**
- [C3] **Wei Li**, Yuzhe Ma, Qi Sun, Yibo Lin, Iris Hui-Ru Jiang, Bei Yu, David Z. Pan, “OpenMPL: An Open Source Layout Decomposer”, IEEE International Conference on ASIC (**ASICON**), Chongqing, China, Oct. 29–Nov. 1, 2019.
- [C2] Xia Li, **Wei Li**, Yuqun Zhang, Yuqun Zhang, Lingming Zhang, “DeepFL: Integrating Multiple Fault Diagnosis Dimensions for Deep Fault Localization”, The ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA**), 2019. **(Distinguished Paper Award)**
- [C1] Bentian Jiang, Xiaopeng Zhang, Ran Chen, Gengjie Chen, Peishan Tu, **Wei Li**, Evangeline F. Y. Young, Bei Yu, “FIT: Fill Insertion Considering Timing”, ACM/IEEE Design Automation Conference (**DAC**), Las Vegas, NV, June 2-6, 2019.

TEACHING EXPERIENCE

Carnegie Mellon University

Head Teaching Assistant

18-765: Digital System Testing and Testable Design **Fall 2025**

- Curated and coordinated a guest seminar series featuring **10+ industry leaders** from **Apple, Broadcom, Qualcomm, Siemens, and Synopsys** to bridge theoretical testing concepts with industrial practices.
- Architected and implemented an **automated grading system**, ensuring consistent feedback for students.
- Led weekly office hours and technical Q&A; coordinated the grading team to synchronize curriculum delivery.

18-202: Mathematical Foundations of Electrical Engineering **Spring 2023**

- Conducted weekly recitations to simplify complex mathematical concepts for students.
- Managed course logistics, grading, and student consultations.

Teaching Assistant

18-765: Digital System Testing and Testable Design **Fall 2022, Fall 2023, Fall 2024**

- Developed and refined course assignments; provided in-depth technical support for projects.

The Chinese University of Hong Kong

Head Teaching Assistant

CENG2030: Fundamentals of Embedded Systems **Spring 2021**

CENG3420: Computer Organization and Design **Spring 2020**

- Designed and supervised weekly laboratory tutorials, focusing on hardware-software co-design and system-level performance.

MENTORING EXPERIENCE

Undergraduate Students Mentored

Aug 2024 - Jan 2026	Yang Zou [C17, 18]	CMU B.S.	UCLA Ph.D.
Aug 2019 - May 2020	Yuxiao Qu [C8]	CUHK B.S.	CMU Ph.D.
Aug 2019 - Dec 2019	Jialu Li [C7]	CUHK B.S.	HKUST M.Phil.
Aug 2019 - Dec 2019	Jialu Xia [C7]	CUHK B.S.	Spring Labs.

Graduate Students Mentored

Sep 2025 - Present	Rebecca Dettmar	CMU M.S.	In progress
Sep 2025 - Present	Zifeng Wang	CMU M.S.	In progress
Sep 2025 - Present	Andrew Kim	CMU M.S.	In progress
Sep 2025 - Jan 2026	Yixin Liang [18]	CMU M.S.	NVIDIA

PROFESSIONAL SERVICE

Technical Program Committee

TPC Member, IEEE International Conference on Computer Design (ICCD) 2026

Journal & Conference Reviewer

- **Journals:** IEEE TCAD, IEEE TIFS, ACM TODAES, The ANZIAM Journal, Integration (the VLSI Journal).
- **Conferences** (Sub-reviewer/External): DAC, ITC, VTS, ETS.

Departmental Service

Student Representative (Volunteer), CMU ECE Faculty Search Committee 2025

- Participated in candidate interviews and social evaluations for Tenure-Track Assistant Professor applicants;
- Provided student-perspective feedback to the hiring committee.

Volunteer, CMU ECE PhD Open House 2025

- Mentored prospective PhD students and introduced research work;

PhD Student Contact 2026

- Provided insights to admitted PhD students about research and life at CMU;
- facilitated their transition into the program.

SELECTED INVITED TALKS & PANELS

2026.09	How AI and LLMs Co-evolve with Hardware Design	SUSTech
2026.08	How AI and LLMs Co-evolve with Hardware Design	HKUST(GZ)
2026.06	How AI and LLMs Co-evolve with Hardware Design	CUHK
2026.03	Towards Autonomous EDA: Research and Vision	Apple
2026.03	Introduction to Graduate Studies: Research Panel	CMU ECE
2026.02	Differentiable ATPG with Mathematical Foundations	Apple
2026.01	Towards Autonomous EDA: Research and Vision	Ricursive Intelligence (Founding Team)
2025.08	When LLMs Meet Graphs	Apple
2025.01	Review on ATPG and Fault Modeling	NVIDIA Research
2024.11	Differentiable Global Routing	Apple